AI/NLP Financial Risk Engine

Turning unstructured news and social text into live risk signals, a sentiment-driven index and event-driven stress tests

Sentiment | Event type | Impact 1-10

Nikhil Chauhan | IIIT Allahabad

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Problem and approach

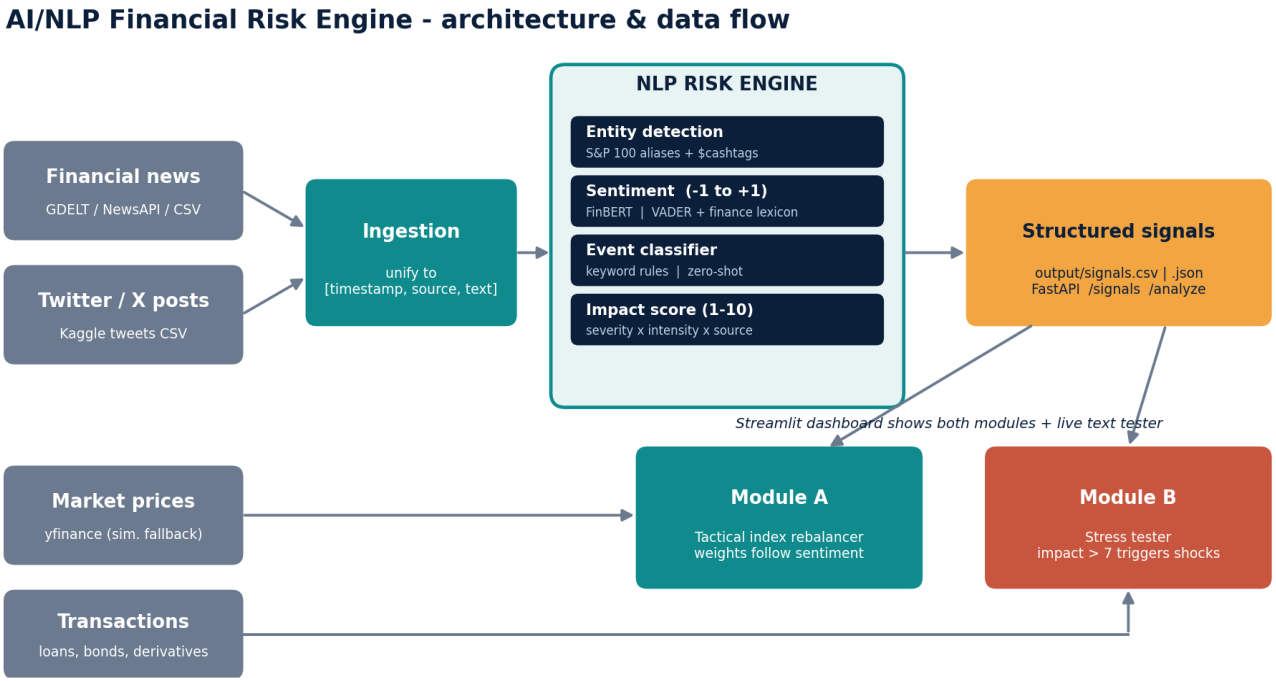
The problem

- Market-moving information arrives as unstructured text: news, tweets, filings
- Analysts cannot read it all, so risk signals arrive late or not at all
- Banks need machine-readable signals that portfolios and risk models can act on

Our approach

1. **Ingest**
   News + Twitter/X posts into one stream
2. **Understand**
   Sentiment, event type and impact score per company
3. **Act**
   Feed signals to a rebalancer and a stress tester

System design and data flow



One engine, many consumers

Signals are plain CSV/JSON plus a FastAPI, so any downstream app can subscribe

Swappable models

FinBERT or VADER for sentiment, rules or zero-shot for events, same output schema

Runs fully offline with sample data; live GDELT and NewsAPI sources are optional.

Implementation highlights

NLP Risk Engine

Entity detection, sentiment in -1 to +1, 7 event types, impact 1-10 with transparent formula and per-signal drivers

Module A: Index rebalancer

Impact-weighted sentiment, smoothed, nudges weights; capped 2-15% so the index stays diversified

Module B: Stress tester

Impact above 7 triggers shocks scaled by severity: equity drop, rate and spread moves, PD uplift, derivative delta

Engineering

Streamlit dashboard, FastAPI, unit tests, labelled eval set, offline fallbacks for every external dependency

Key results

84

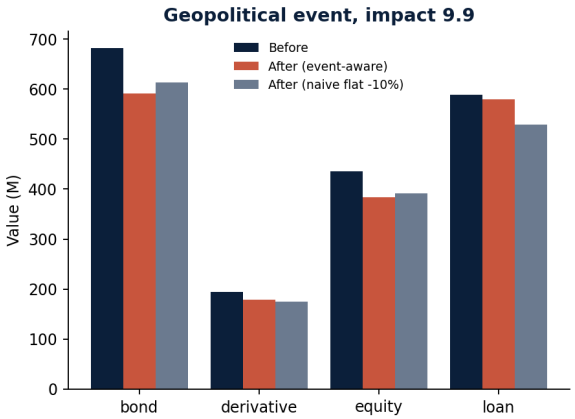
signals from 2 sources

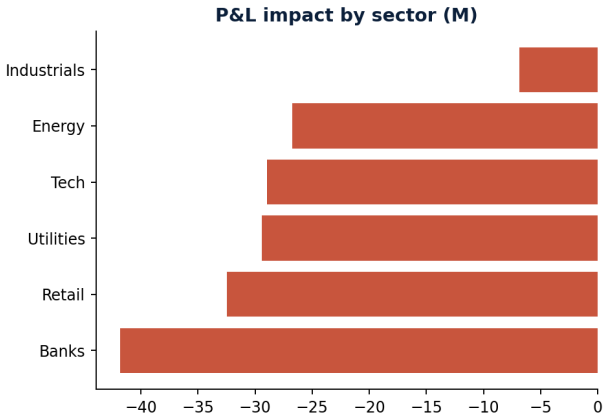
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high-impact events (score > 7)

-8.76%

portfolio value after a Geopolitical shock





Event-aware stress test: 166.3M loss on a 1,899M portfolio, concentrated in bonds (90.6M) and equities (51.7M). A flat -10% shock would say 189.9M and badly overstate the loan loss (8.5M here).

Module A: sentiment moved WMT up 6.4pp and GOOGL down 4.7pp within the 2-15% weight caps.

Synthetic sample data and simulated prices for the offline demo run; hand-labelled test set is small (n=41), so accuracy is indicative only.

Domain impact

1

Risk and credit teams

Early, quantified warning when a news event threatens exposures, instead of discovering it from price moves

2

Portfolio managers

Systematic, explainable tilts from text sentiment, with weight caps that protect diversification

3

Wholesale banking and treasury

Instant what-if on loans, bonds and derivatives when a geopolitical or credit event hits, by asset class and sector

From reading headlines to a machine-readable risk signal in milliseconds

Limitations and next steps

Limitations

- Rule-based events and fallback lexicon miss nuance and sarcasm
- Impact score is heuristic, not calibrated to realised market moves
- Stress shocks are simplified (duration and PD multipliers)
- Small self-labelled evaluation set

Next steps

- Switch on FinBERT and zero-shot events (already supported)
- Calibrate impact against historical returns; track the information coefficient
- Stream with Kafka and add dedup and source credibility
- Historical-scenario and VaR-based stress library